

AWS Route 53 Complete Study Guide

Step-by-Step Public Hosted Zone Creation, DNS Registrar Delegation, A Record Setup & In-Depth Routing Policies



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AWS Route 53 Architecture & Hands-on Guide

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1. Introduction to AWS Route 53

Amazon Route 53 is a highly available, scalable, and fully managed Domain Name System (DNS) web service provided by AWS. It is designed to give developers and businesses an extremely reliable and cost-effective way to route end users to Internet applications by translating human-readable names (like `www.example.com`) into numeric IP addresses (like `192.0.2.1`).

Key Core Functions of Route 53

- **Domain Registration:** Register and manage domain names directly within AWS.
- **DNS Routing:** Route user requests to infrastructure such as EC2 instances, CloudFront distributions, S3 buckets, or external endpoints.
- **Health Checking:** Monitor the health of resources and automatically failover traffic when endpoints are unhealthy.

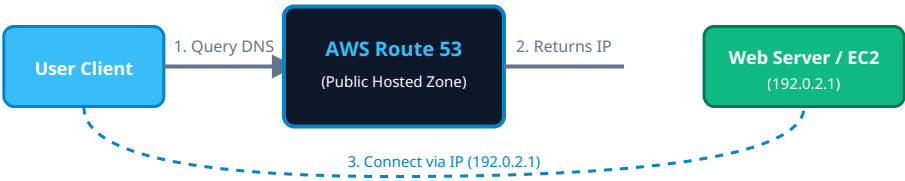


Figure 1: High-Level Overview of Route 53 DNS Resolution Flow. (Prepared by Prabu Thiagamani)

2. Essential DNS & Route 53 Concepts

Term	Description	Example / Usage
Hosted Zone	A container for DNS records that define how to route traffic for a specific domain.	Public Hosted Zone for <code>mycompany.com</code> .
A Record	Maps a domain/subdomain name directly to an IPv4 address.	<code>app.mycompany.com</code> → <code>192.0.2.10</code>
AAAA Record	Maps a domain/subdomain name directly to an IPv6 address.	<code>app.mycompany.com</code> → <code>2001:db8::1</code>
CNAME Record	Maps a domain name to another domain name (Canonical Name). Cannot be used at Zone Apex.	<code>blog.mycompany.com</code> → <code>web.mycompany.com</code>
Alias Record	AWS-specific extension to DNS. Points to AWS resources (CloudFront, ALB, S3). Works at Zone Apex.	<code>mycompany.com</code> → <code>d123.cloudfront.net</code>
NS Record	Name Server record. Specifies which DNS servers are authoritative for the domain.	<code>ns-123.awsdns-15.com</code>
SOA Record	Start of Authority. Contains administrative and technical information about the domain zone.	Primary NS, contact email, zone refresh timings.

■ 3. Step-by-Step Guide: Creating a Public Hosted Zone & Configuring DNS

Follow these exact steps to host your domain's DNS in AWS Route 53, update your third-party domain registrar, and route web traffic to your web server IP address.

Step 1: Create a Public Hosted Zone in AWS Route 53

1. Sign in to the **AWS Management Console** and open the **Route 53 Console**.
2. In the left navigation pane, click on **Hosted zones**.
3. Click the blue **Create hosted zone** button.
4. In the **Domain name** field, enter your registered domain name (e.g., `exampledomain.com`).
5. Select **Public hosted zone** under the *Type* section.
6. (Optional) Add tags (e.g., `Environment = Production`) for management.
7. Click **Create hosted zone**. Route 53 automatically creates an **NS** and an **SOA** record set.

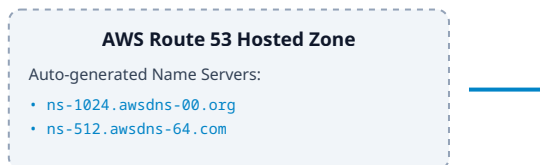


Figure 2: Transferring Name Server (NS) Authority to AWS Route 53. (Author: Prabu Thiagamani)

Step 2: Configure NS Records in Your Domain Name Registrar

Open your newly created Hosted Zone in the AWS Route 53 Console and locate the record of type **NS**.

Copy the **4 Name Server endpoints** listed in the Value column (e.g., `ns-xxx.awsdns-xx.com`).

Log in to your third-party domain registrar portal (e.g., GoDaddy, Namecheap, Cloudflare, Name.com).

Navigate to the **DNS Management** or **Nameservers** section for your domain.

Change the Nameserver setting from "Default" to "**Custom Nameservers**".

Paste all 4 AWS Route 53 name server addresses into the custom nameserver fields (remove trailing dots if present).

Save changes. *Note: DNS delegation propagation can take anywhere from a few minutes to 24–48 hours worldwide.*

Step 3: Add an A Record in AWS Route 53

1. Return to the **Route 53 Console** → **Hosted zones** → Click your domain name.
2. Click the orange **Create record** button.
3. In the **Record name** field, enter a subdomain (e.g., `www`) or leave it blank for the Apex domain (`exampledomain.com`).
4. In the **Record type** dropdown, choose **A - Routes traffic to an IPv4 address and some AWS resources**.
5. Toggle **Alias** switch to *Off* if routing to a standard server IP address.
6. In the **Value** box, enter your public IPv4 address (e.g., `203.0.113.25`).
7. Set the **TTL (Time To Live)** (Default: 300 seconds).
8. Select **Simple routing** under Routing policy.
9. Click **Create records**. Your A record is now live!

■ 4. In-Depth Guide to AWS Route 53 Routing Policies

AWS Route 53 offers flexible routing policies to determine how DNS queries are responded to based on performance, geography, latency, and availability.

1. Simple Routing Policy

Use Case: Single-resource domain, or returning multiple random values for basic load distribution.

Routes traffic to a single resource (e.g., a single web server). You can specify multiple values in a single record (e.g., multiple IP addresses); Route 53 returns all values to the client in a randomized order.

2. Weighted Routing Policy

Use Case: A/B testing, blue/green software deployments, or gradual traffic migration.

Assigns relative weights (e.g., 80 to Region A, 20 to Region B) to different endpoints. Route 53 calculates the proportion of traffic sent to each resource based on the assigned weights relative to the total sum.

3. Latency-Based Routing Policy

Use Case: Global applications requiring minimum latency for users across multiple AWS regions.

Route 53 routes client queries to the AWS region that provides the lowest network latency for the requesting user. AWS continuously measures latency between clients and data centers worldwide.

4. Failover Routing Policy

Use Case: Active-Passive Disaster Recovery setup.

Configures a primary active site and a secondary passive disaster recovery site. Coupled with Route 53 Health Checks, queries automatically redirect to the secondary site if the primary site becomes unhealthy.

5. Geolocation Routing Policy

Use Case: Localizing content, restricting distribution based on legal/compliance requirements.

Routes traffic based on the geographic location of the user making the request (by Continent, Country, or US State). Allows customized language pages or regional content restrictions.

6. Geoproximity Routing Policy (Traffic Flow)

Use Case: Dynamic shifting of traffic boundaries between multiple geographical regions.

Routes traffic based on the geographic location of users and resources. You can optionally specify a *bias* value to expand or shrink the geographic region from which traffic is routed to a specific resource.

7. Multi-Value Answer Routing Policy

Use Case: Simple DNS-based load balancing with health checks.

Returns up to 8 healthy IP records in response to DNS queries. Unlike simple routing, Multi-Value Answer integrates directly with Route 53 health checks to return *only healthy* endpoints to clients.

Routing Policy Comparison Matrix

Routing Policy	Primary Decision Factor	Supports Health Checks?	Primary Application Target
Simple	Static single endpoint or random list	No	Single web server / static site
Weighted	Configured relative percentage (%)	Yes	Canary testing, soft rollouts
Latency	Lowest network response time	Yes	Global multi-region performance
Failover	Health Check Status (Active / Passive)	Yes	Disaster recovery failover
Geolocation	User's physical location (Country/State)	Yes	Localized content & compliance
Geoproximity	Distance + Regional Bias adjustment	Yes	Complex global traffic steering
Multi-Value	Health check filtered random picks	Yes	Client-side load balancing